

Project Specification

Introduction

Your task is to create a DSL. Try to be creative. Your DSL can compile to any language of your choice. In the class demo, the DSL compiles to Python but you are free to compile to any language of your choice HTML, Python, Assembly, etc. The choice is limitless. Just try to be creative.

What is a DSL?

- We have seen general-purpose languages (GPLs) such as Java, C++, Python.
- A language with a higher level of abstraction optimized for a specific class of problems.
- A DSL uses the concepts and rules from the field or domain.
- A DSL uses the concepts and rules from the field or domain while sacrificing some of the flexibility to express any program.
- Examples:
 - [Mermaid JS](#)
 - [Jinja](#)
 - Google sheets
 - JUnit

How much will the project contribute towards the overall grade?

The project will contribute to **40%** of the overall lab grade. The assignments, lab tests etc. will contribute to the remaining **60%**.

What is the deadline?

- The project deadline is **13-April-2026** end of day. This is a hard deadline and no extensions are allowed. The presentation of the demos are scheduled for 14, 15, 16, 17 April with randomly chosen teams on each day. More information on this in coming weeks.
- You can submit after the deadline with penalty as mentioned later.
- There will be no assignment on **07-April-2026** but you must come to the lab and we expect an initial demo by this day. You can show us the demo if you want and ask questions about the project. We will not grade the demos on this day.

Any penalties for not using OCaml?

No. Although we recommend OCaml, you are free to choose any programming language for your implementation. There are be no penalties for this.

Is there a penalty for missing the deadline for the project submission?

Yes, will be decided case-case basis by the instructor. It can be as high as **75%** of overall grade for the project.

What are the final deliverables?

Each team must submit only once. Please choose one person from your team as the **team leader**. The only task of the team leader is to make a project submission. Multiple submissions are not allowed and may be penalized. Make sure you submit only **one** final version. We will float a google form for the submission.

- A **private** GitHub repo link containing all code and README. (We will check your GitHub and if you make any commits after the deadline, you will be penalized)
- An **unlisted** YouTube video link. (You must not add any new features after the deadline. Any added features after the deadline will be penalized.)

What if I do not perform an user survey?

Although the user survey part has a small component in the objective grade, not doing an user survey will deduct up to 20% from the overall final grade of the project.

Team

What is the size of a Team?

Each team will have four team members. We have created the four member teams for you. There will be few teams with five members.

I prefer to work alone, can I create my own team?

Short answer is **No**. Software development is a team effort. In real world, you will be part of a team.

Is there penalty for different team sizes?

No. All team sizes are graded equally. We **will check** for contributions from the team participants based on what you submit and interactions during the demo.

Will all team members be graded the same?

No. We will check the submitted GitHub repo for contributions by individual team members. Write the estimated contributions by individual team members explicit in your README of the repo.

Grading Scheme

Subjective Evaluation [30]

1. Innovation in DSL (29)
 - Marks by TAs.
 - Marks by instructors.
2. Team dynamics **perception** during demo (1)

Objective Evaluation [70]

1. Have a **private** repo in GitHub. Add TAs as contributors. You are free to make the repo public after the end semester results are announced. We strictly prohibit it to make it public before.
2. Integrated GitHub actions in the repo. **(10)**
 - Integrated linting in GitHub actions (If you are using OCaml, linters are available such as [Merlint](#), [Zanadu](#))
 - Integrated running tests in GitHub actions. (If OCaml, use OUnit2 for this integration. For other language you should find something yourself and we can't help you with this).
 - You should have at least ten distinct test programs (each program should be more than five lines) written by your DSL. The idea is to check for normal case and the corner cases.
 - For each function implemented (unless it is not possible to write unit tests) by you, you should have at least three unit tests (this is the bare minimum, you can of course have more).
3. Have followed the specification of the DSL and have implemented at least **three** of the following features. More features are welcome. **No part marks for this. (15)**
 - Mutable variable (variable whose value can be changed).
 - Loops and/or Recursions.

- Conditions.
 - Complex data types (lists, arrays, ...).
 - Functions.
4. Designed an User Study **(10)**
 - Collect feedback from user study.
 - Implemented some weakness pointed out by the user study. Not *all* feedback needs to be incorporated but at least a few should be there. Mentioning your DSL does not have any weakness is not acceptable.
 5. Documentation **(20)**

The project is properly documented in *README.md*.
 6. Team Contribution **(10)**.
 - Each of you **must** contribute to the code present in the repo.
 - Design/Documentation.
 7. 5 min video demo of the project to be uploaded (unlisted YouTube link). **(5)**

User Survey

The **goal** of user survey is to identify an user base for your DSL and get some initial feedback. Ideally, you should schedule your user survey once you have created an initial prototype of the DSL. You should keep some time to implement some of the feedback obtained during the survey.

For example, if your DSL generates a game for non-programmers, identify such a target audience, ask them to use and interact with the DSL and ask questions. Preparing good questions is the key. Simply asking

- "Is my DSL good?" Yes/No
is probably not a good feedback that you can use to improve your design.

The user survey should have a minimum number of two participants/users. This is bare minimum but there is no upper bound on the number of participants.

The following can't be a participant of the user study

1. The members from the same team can't participate.
 2. The TAs. Any feedback from the TA can't be counted towards a user feedback.
 3. The instructors. Any feedback from the instructor can't be counted towards a user feedback.
- Apart from the above, there is no other restrictions.

Ideas

1. How to design a good User Survey? You may check this [YouTube](#) video to get some ideas. Free to search more.
2. Need to have at least five questions (this number is bare minimum). Collect feedback from the users. You may ask about the strengths and weakness of your DSL etc.

README

The *README.md* should be in your GitHub repo. You do not need to submit it separately.

The document should have at least the following sections

The following is the bare minimum you are free to add more.

- A brief description about the project that includes motivation for the DSL. Who is the target user group of the DSL?
- A detailed description about the DSL eg. the keywords, allowed characters etc.
- One sample program written by your DSL, and a corresponding sample final output that gets generated. For example, if the final output is some UI, you can put a screenshots/embed some short video.
- Clear detailed description of the member contribution. Put each team member's name, roll number and one paragraph about the contribution. Also put the percent (%) of overall effort by the team member towards the overall project.
- Documentation on how to build and run your project? How to run the tests?
- What questions were asked in the user survey? A summary of the responses by the users.
- Unlisted 5 minute YouTube link of the project demo.